

Key Analytical Findings & SQL Queries

1. Global Total Debt

- **Question:** What is the total amount of debt owed by all countries in the dataset?
- **Answer & Finding:** The total external debt accumulated across all valid country records is **\$2,823,893,300,273.1 USD** (~\$2.82 Trillion).

SQL

```
SELECT  
  
    ROUND(SUM(debt)::NUMERIC, 2) AS total_debt_usd  
  
FROM international_debt_with_missing_values  
  
WHERE debt IS NOT NULL;
```

2. Distinct Countries Count

- **Question:** How many distinct countries are recorded in the dataset?
- **Answer & Finding:** There are **124** distinct countries recorded in the dataset.

SQL

```
SELECT  
  
    COUNT(DISTINCT country_name) AS distinct_country_count  
  
FROM international_debt_with_missing_values  
  
WHERE country_name IS NOT NULL AND TRIM(country_name) != '';
```

3. Distinct Debt Indicators

- **Question:** What are the distinct types of debt indicators, and what do they represent?
- **Answer & Finding:** The dataset tracks **25** distinct debt indicators covering bilateral, multilateral, official creditor, and private nonguaranteed loans.

SQL

```
SELECT DISTINCT  
  
    indicator_code,  
  
    indicator_name  
  
FROM international_debt_with_missing_values  
  
WHERE indicator_code IS NOT NULL AND TRIM(indicator_code) != ''  
  
    AND indicator_name IS NOT NULL AND TRIM(indicator_name) != ''  
  
ORDER BY indicator_code;
```

4. Highest Debt-Holding Country

- **Question:** Which country has the highest total debt, and how much does it owe?
- **Answer & Finding:** **China** holds the highest individual debt obligation at **\$283,748,948,517.90 USD** (~\$283.75 Billion), followed by **Brazil** (\$274.09B), regional aggregate **South Asia** (\$243.69B), **Least Developed Countries (UN)** (\$167.51B), and **Russian Federation** (\$167.39B).

SQL

```
SELECT
    country_name,
    country_code,
    ROUND(SUM(debt)::NUMERIC, 2) AS total_debt_usd
FROM international_debt_with_missing_values
WHERE country_name IS NOT NULL
    AND TRIM(country_name) != ''
    AND debt IS NOT NULL
GROUP BY country_name, country_code
ORDER BY total_debt_usd DESC NULLS LAST
LIMIT 5;
```

5. Average Debt Across Indicators

- **Question:** What is the average debt across different debt indicators?
- **Answer & Finding:** Long-term principal repayments (DT.AMT.DLXF.CD) average the highest single obligation per record at **\$6,467,164,394.26 USD** (~\$6.47 Billion), demonstrating that long-term debt servicing accounts for the largest capital outflows.

SQL

```
SELECT
    indicator_code,
    indicator_name,
    ROUND(AVG(debt)::NUMERIC, 2) AS average_debt_usd,
    COUNT(*) AS record_count
FROM international_debt_with_missing_values
WHERE indicator_code IS NOT NULL AND TRIM(indicator_code) != ''
    AND indicator_name IS NOT NULL AND TRIM(indicator_name) != ''
    AND debt IS NOT NULL
GROUP BY indicator_code, indicator_name
ORDER BY average_debt_usd DESC NULLS LAST;
```

6. Principal Repayments Leaderboard

- **Question:** Which country has made the highest amount of principal repayments?
- **Answer & Finding:** **China** has made the highest total principal repayments at **\$168,611,607,050.00 USD** (~\$168.61 Billion), followed by **Brazil** (\$131.87B), the **Russian Federation** (\$109.39B), **Turkey** (\$91.47B), and **South Asia** (\$73.13B).

SQL

```
SELECT
    country_name,
    ROUND(SUM(debt)::NUMERIC, 2) AS total_principal_repayments_usd
FROM international_debt_with_missing_values
WHERE indicator_name ILIKE '%principal repayments%'
    AND country_name IS NOT NULL
    AND TRIM(country_name) != ''
    AND debt IS NOT NULL
GROUP BY country_name
ORDER BY total_principal_repayments_usd DESC NULLS LAST
LIMIT 10;
```

7. Most Common Debt Indicator

- **Question:** What is the most common debt indicator across all countries?
- **Answer & Finding:** The most frequent debt indicator across country reporting profiles is **PPG, official creditors (INT, current US\$)** (DT.INT.OFFT.CD), appearing across 107 country records.

SQL

```
SELECT
    indicator_name,
    indicator_code,
    COUNT(*) AS indicator_frequency
FROM international_debt_with_missing_values
WHERE indicator_name IS NOT NULL AND TRIM(indicator_name) != ''
GROUP BY indicator_name, indicator_code
ORDER BY indicator_frequency DESC
LIMIT 5;
```

8. Key Debt Trends & Macro Disparity

- **Question:** Identify any other key debt trends and summarize your findings[cite: 11].
- **Answer & Finding:** Debt distribution is heavily skewed. The top 10 debt holders account for over **50% of global debt obligations**, whereas small developing states (such as *Sao Tome and Principe* at \$44.80M, *Tonga* at \$55.21M, and *Comoros* at \$57.57M) hold less than 0.005% of global debt.

SQL

-- Top 10 Debtors

SELECT

country_name,

ROUND(SUM(debt)::NUMERIC, 2) AS total_debt_usd

FROM international_debt_with_missing_values

WHERE country_name IS NOT NULL AND TRIM(country_name) != '' AND debt IS NOT NULL

GROUP BY country_name

ORDER BY total_debt_usd DESC NULLS LAST

LIMIT 10;

-- Least Indebted Nations

SELECT

country_name,

ROUND(SUM(debt)::NUMERIC, 2) AS total_debt_usd

FROM international_debt_with_missing_values

WHERE country_name IS NOT NULL AND TRIM(country_name) != '' AND debt IS NOT NULL

GROUP BY country_name

ORDER BY total_debt_usd ASC NULLS LAST

LIMIT 10;